

AYUSH BARDHANI

Software Engineer | Backend & AI/ML

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SUMMARY

Software engineer focused on developing full-stack applications, secure REST APIs, and production-deployed systems across Node.js/Express and Django. Applies Python-based machine learning and deep learning (scikit-learn, TensorFlow) alongside LLM APIs (OpenAI, Gemini) to ship AI-powered features, with strong DSA fundamentals in Java and working knowledge of AWS cloud architecture and system design.

PROJECTS

ScrowX — Secure Payment Infrastructure Layer For Informal Transactions

React 19 (Vite), Node.js, Express.js, Socket.IO, PostgreSQL, Prisma, Razorpay, JWT, Google Gemini API, xAI Grok API

[GitHub](#) | [Live](#) | [Demo](#)

- Architected a secure payment infrastructure layer with an escrow engine (Node.js/Express, Razorpay), enforcing atomic fund lock/release via Prisma transactions and row-level locking to eliminate double-spend and race conditions.
- Designed an append-only wallet ledger where balances are always derived from transaction aggregates rather than a mutable field, ensuring tamper-evident, audit-ready financial history.
- Built an AI-powered dispute arbitration engine (Google Gemini) that analyzes evidence and behavioral risk signals, with multi-round challenge logic and auto-escalation to human review for high-value/low-confidence cases.
- Deployed a full-stack production system: React/Vite (Netlify), Express + Socket.IO (Render), PostgreSQL/Prisma, JWT auth with KYC gating, real-time chat, and an AI support assistant (xAI Grok, OpenAI fallback).

AWS Cloud Cost Management System

Python, FastAPI, React (Vite), scikit-learn, Boto3, JWT, Docker, APScheduler, SQLite

[GitHub](#)

- Built a full-stack finance tool to track live AWS usage/spend across EC2, S3, RDS, and Lambda via Boto3, with automated cost breakdowns; implemented JWT auth with Fernet-encrypted credential storage for isolated multi-account monitoring.
- Built a cost-forecasting module (scikit-learn polynomial regression, R²-validated) and CloudWatch-based idle-resource detection with cost-ranked cleanup recommendations refreshed every 6 hours via APScheduler; containerized the backend with Docker.

TrackAI — Railway Traffic Data Pipeline & Delay Prediction

Python, Pandas, NumPy, scikit-learn, Matplotlib

[GitHub](#)

- Built an end-to-end data pipeline to clean, engineer features from, and model large-scale railway traffic data.
- Trained delay-prediction models (85%+ accuracy) via k-fold cross-validation and hyperparameter tuning; built Matplotlib dashboards surfacing congestion hotspots and delay trends for timetable optimization.

ML Notebooks

- **CreditWise — Loan Approval Prediction (Supervised):** compared Logistic Regression, KNN, and Naive Bayes on 1,000 loan applicants; best model reached 87.5% accuracy and 0.80 F1-score after feature engineering. [GitHub](#)
- **SmartCart — Customer Segmentation (Unsupervised):** applied PCA and K-Means/Agglomerative clustering (elbow method + silhouette scoring) to segment 2,240 e-commerce customers into 4 actionable groups by income, spend, and channel behavior. [GitHub](#)

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), Python, SQL

Backend & Frameworks: Node.js, Express.js, Django, REST APIs, WebSockets, JWT Authentication, RBAC & Permissions, Prisma ORM

Databases: PostgreSQL, MongoDB, MySQL, SQLite

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda), Boto3, Docker, GitHub Actions (CI/CD), Git/GitHub, Railway, Vercel, Netlify, Render.

Applied AI / ML: OpenAI API, Google Gemini API, scikit-learn, TensorFlow, Machine Learning (LR, KNN, SVM, Decision Tree, Random Forest etc), Deep Learning (ANN, FNN, CNN, RNN), Pandas, NumPy, Web Scraping, Data Visualization(Matplotlib, Seaborn), Feature Engineering.

Core CS: Data Structures & Algorithms (Java), OOP, DBMS, Operating Systems, Computer Networks, System Design

CERTIFICATIONS & TRAINING

AWS Academy: Cloud Architecting, Cloud Foundations, Introduction to Cloud Semester

Cisco: Introduction to Cybersecurity, Industrial Networking Essentials, Apply AI

ACHIEVEMENTS

- Solved 200+ DSA problems (Java) spanning arrays, linked lists, trees, graphs, sliding window, recursion, and dynamic programming.
- Finalist, Stellaris — MLH-affiliated hackathon by GeeksforGeeks (GFG); built a dispute-resolution system for freelance transactions.

EDUCATION

ABES Engineering College, Delhi, India Oct 2023 – July 2027

B.Tech, Computer Science & Engineering — Artificial Intelligence & Machine Learning | CGPA: 7.64 / 10

LEADERSHIP & EXTRACURRICULAR

Promotional Head, Trishul ABESEC 2023 – 2026

- Organized 15+ campus events and awareness campaigns engaging 200+ students, leading promotional strategy for a 40+ member team.